

TJAKAAPE, Sweden

WMO: 021410

Lat: 66.300N

Lon: 19.200E

Elev: 582

StdP: 94.53

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-22.6	-27.7	0.3	-25.0	-24.9	0.4	-22.4	9.1	-2.0	8.1	-3.5	0.5	20	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	23.4	14.8	21.3	13.8	19.3	13.0	16.3	21.1	15.1	19.2	14.0	17.5	3.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	11.2	17.2	13.4	10.3	16.1	12.4	9.6	15.0	47.8	21.3	44.1	19.0	40.9	17.8	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	6.1	-28.8	26.5	2.9	2.4	-30.9	28.2	-32.5	29.6	-34.1	31.0	-36.2	32.8		
(5)			-29.1	17.9	2.5	1.6	-30.9	19.0	-32.4	19.9	-33.8	20.8	-35.6	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-11.2	-11.0	-7.0	-1.6	4.4	9.5	13.0	10.8	5.9	-1.0	-6.4	-8.6
	DBStd	9.58	6.09	6.25	4.97	4.14	4.16	3.89	3.26	3.32	3.34	4.50	4.86	5.56
	HDD10.0	3935	659	587	527	348	182	56	7	29	127	341	493	578
	HDD18.3	6774	917	821	786	598	432	267	168	234	373	599	743	836
	CDD10.0	208	0	0	0	0	8	40	101	54	4	0	0	0
	CDD18.3	5	0	0	0	0	0	1	4	1	0	0	0	0
	CDH23.3	61	0	0	0	0	1	13	41	5	0	0	0	0
	CDH26.7	6	0	0	0	0	0	1	4	0	0	0	0	0
Wind	WSAvg	2.7	1.5	2.2	3.2	3.1	3.2	3.3	3.0	2.8	3.0	2.8	2.1	2.0
Precipitation	PrecAvg	605	39	27	28	26	46	70	99	75	55	48	46	41
	PrecMax	846	83	51	57	56	95	127	147	146	188	100	94	125
	PrecMin	370	1	1	7	7	11	23	28	25	16	16	12	9
	PrecStd	118	18	16	14	15	23	28	32	36	37	21	25	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	3.9	5.9	12.6	21.8	25.1	26.9	24.0	17.4	9.7	5.3	3.1
		MCWB	2.8	2.5	2.6	6.4	11.2	15.2	17.5	15.4	12.0	7.5	4.6	2.2
	2%	DB	1.0	1.8	3.3	9.1	18.3	21.8	24.3	21.7	14.5	7.9	3.4	1.5
		MCWB	0.1	0.4	0.9	4.1	10.3	13.0	15.4	14.7	10.8	6.4	2.9	0.8
	5%	DB	-1.2	-0.3	1.7	6.5	14.6	19.4	22.1	19.5	12.5	6.6	1.3	0.0
		MCWB	-1.6	-1.2	-0.1	2.8	8.7	11.8	14.9	13.8	9.7	5.5	1.0	-0.3
	10%	DB	-3.3	-2.1	0.3	4.7	11.7	16.9	19.9	17.2	11.2	5.1	0.0	-1.4
		MCWB	-3.5	-2.6	-1.1	1.9	7.1	11.0	13.7	13.0	9.1	4.4	-0.2	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.5	2.7	6.5	12.5	16.5	18.9	17.3	12.8	8.3	4.7	2.4
		MCDB	3.9	3.7	4.7	11.9	19.5	22.2	23.6	21.2	14.2	8.8	4.9	3.0
	2%	WB	0.4	0.6	1.2	4.7	11.0	14.6	16.8	15.7	11.5	6.7	2.9	0.8
		MCDB	0.9	1.5	2.9	8.1	16.4	19.3	22.3	19.2	13.6	7.5	3.3	1.3
	5%	WB	-1.6	-1.1	0.2	3.3	9.0	13.0	15.6	14.7	10.5	5.5	1.0	-0.3
		MCDB	-1.2	-0.4	1.6	5.8	14.4	17.4	20.3	18.0	12.0	6.4	1.3	0.0
	10%	WB	-3.5	-2.7	-0.9	2.1	7.3	11.5	14.4	13.5	9.6	4.4	-0.2	-1.6
		MCDB	-3.3	-2.1	0.2	4.6	11.0	16.0	18.4	16.5	10.7	5.1	0.0	-1.4
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.1	8.1	8.3	8.4	9.0	9.1	8.9	7.2	5.6	5.4	5.8
		MCDBR	6.4	6.5	7.7	10.3	13.8	12.7	12.9	12.8	10.1	6.6	4.9	5.5
		MCWBR	6.2	6.0	6.1	6.4	7.4	6.0	5.9	6.4	6.4	5.1	4.7	5.4
	5% WB	MCDBR	6.2	6.7	7.4	9.0	13.2	10.9	11.2	10.5	7.8	6.0	4.8	5.5
		MCWBR	6.1	6.3	6.1	5.8	7.3	5.8	5.7	6.0	5.3	4.9	4.7	5.4
Clear-Sky Solar Irradiance	taub	0.172	0.240	0.273	0.301	0.318	0.320	0.330	0.315	0.287	0.265	0.340	0.175	
	taud	1.939	2.174	2.245	2.305	2.488	2.527	2.509	2.535	2.592	2.416	2.838	1.954	
	Ebn at Noon	271	637	797	865	888	893	871	850	803	619	291	33	
	Edh at Noon	16	50	78	96	90	88	77	57	40	17	3		
All-Sky Solar Radiation	RadAvg	0.08	0.60	2.06	3.88	4.92	5.11	4.71	3.53	1.97	0.71	0.15	0.01	
	RadStd	0.01	0.09	0.13	0.20	0.49	0.47	0.53	0.35	0.26	0.12	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page